



Step-by-Step: Verification of data cleaning

This reading outlines the steps the instructor performs in the following video, [Verification of data cleaning](#). The video demonstrates how to verify cleaned data in both spreadsheets and SQL.

Keep this step-by-step guide open as you watch the video. It can serve as a helpful reference if you need additional context or clarification while following the video steps. This is not a graded activity, but you can complete these steps to practice the skills demonstrated in the video.

What you'll need

If you'd like to follow along with the examples in this video, choose a spreadsheet tool. Google Sheets or Excel are recommended.

To access the spreadsheet the instructor uses in this video, click the link to the template to create a copy of the dataset. If you don't have a Google account, download the data directly from the attachments below.

Link to dataset: [Jeff's Party Planet - Data for Cleaning](#)

OR

Note: The SQL table used in this example is not available for this activity.

Example 1: Verify data with spreadsheets

Use spreadsheet tools such as Find and Replace and pivot tables to find, understand, and fix errors in your spreadsheet.

Use Find and Replace to replace all instances of a mistake

1. Use the [Jeff's Party Planet - Data for Cleaning](#) dataset.
2. From the Edit menu, choose Find and Replace to open the Find and replace dialog box.
3. In the Find field, enter the misspelled word in the supplier name, Plos.
4. In the Replace with field, enter Plus.
5. Click Replace all to replace all instances of "Plos" with "Plus". Click Done to close the Find and replace dialog box.
6. Select the Undo button to use a different method to correct this misspelling. This can also be done with Ctrl (Windows) or Command (Mac) Z.

Use a pivot table to understand errors in a spreadsheet

1. Select the Suppliers column.
2. Select Insert > Pivot Table. In the Create pivot table dialog box, choose New Sheet then Create.
3. This creates a new tab that is mostly blank.
4. Additionally, the Pivot table editor pane is in the window.
5. Next to Rows. Select Add, then the Suppliers column.
6. Next to Values, select Add then select Suppliers. This adds a value for the Suppliers column.
7. By default, Google Sheets sets the value to summarize by **COUNTA** (the total number of values in a range). This will show how many times each supplier name comes up. It's a great way to check for misspellings and other anomalies. Note: Don't use COUNT, because COUNT counts only numerical values.

8. When there is only one instance of the misspelled name, manually change it to the correct spelling.
9. To return to the original sheet, select the Sheet1 tab.

Example 2: Use a CASE statement to verify data in SQL

Use **CASE** statements to correct misspellings in SQL.

1. The SQL table used in this example is not available for download, but if you were performing a similar query you'd first make sure to load the data in BigQuery.
2. Start your SQL query with the basic structure:

SELECT

FROM

WHERE

3. In the **FROM** clause, specify the table you're pulling data from after **FROM**. For example, **project-id.customer_data.customer_name**
4. In the **SELECT** clause, specify the columns you want to return. In this example, you want **customer_id** and **first_name**.
5. However, there is a misspelling in a customer's first name.
 - i. To correct the misspelled name "Tnoy" to "Tony", use a **CASE** statement.
 - ii. Enter **CASE**. On the next line, enter **WHEN first_name = 'Tnoy' THEN 'Tony'**. This tells SQL to replace any instances of **Tnoy** in the **first_name** column with **Tony**.
 - iii. On the next line, add the **statement ELSE first_name** to keep other names as they are.

iv. End the statement with `END AS cleaned_name`. This creates a new column called `cleaned_name` that will contain the data cleaned with the `CASE` statement.

6. Delete the `WHERE` clause because you don't want to filter the query.

7. The final statement should be:

```
SELECT
```

```
    Customer_id,
```

```
    CASE
```

```
    WHEN first_name = 'Tnoy' THEN 'Tony'
```

```
    ELSE first_name
```

```
    END AS cleaned_name
```

```
FROM
```

```
    project-id.customer_data.customer_name
```

8. This SQL query will correct the misspelled name and leave other names unchanged in a new column called `cleaned_name`. Note that this query corrects only the display of the name; it does not update the table's data.

Key Concepts:

- **CASE Statement:** A SQL construct that allows for conditional logic in queries, enabling the correction of specific data entries based on defined criteria.
- **Pivot Table:** A data processing tool in spreadsheets that summarizes data, helping to identify errors and anomalies in datasets.
- **Find and Replace:** A spreadsheet tool that allows users to search for specific text and replace it with new text, useful for correcting errors.